Reading Notes of Chari, Christiano, & Kehoe (1994) "Optimal Fiscal Policy in a Business Cycle Model"

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1 Decentralized Economy

In each period $t = 0, 1, \ldots$, the economy experiences one of finitely many events s_t. Denote $s^t \triangleq (s_0, \ldots, s_t)$ as the history of events up through and including period t, and $\mu(s^t)$ as the unconditional probability of any particular history s^t. The initial realization s_0 is given.

The technology is neoclassical:

$$y_t = F(k_t, n_t, s_t), \tag{1.1}$$

where $k_t \equiv k(s^{t-1})$ is capital and $n_t \equiv n(s^t)$ is labor.

Competitive pricing implies

$$r_t = F_k(k_t, n_t, s_t), \tag{1.2}$$
$$w_t = F_n(k_t, n_t, s_t), \tag{1.3}$$

where $r_t \equiv r(s^t)$ and $w_t \equiv w(s^t)$ refer to before-tax returns on capital and labor, respectively.

The output can be used for private consumption $c_t \equiv c(s^t)$, exogenous government consumption $g_t \equiv g(s^t)$, and saving k_{t+1}.

The preference of the representative agent is given by

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu(s^t) U(c_t, n_t), \tag{1.4}$$

where U satisfies standard assumptions.

The government is allowed to levy linear taxes on the income from labor and capital (lump-sum tax is forbidden), and to issue a short-term debt. Let $\tau_t^n \equiv \tau^n(s^t)$ and $\tau_t^k \equiv \tau^k(s^t)$ denote the tax rates on the income from labor and capital. Let $b_{t+1} \equiv b(s^t)$ denote the number of units of debt issued at state s^t, and $R_t^b \equiv R^b(s^{t+1})$ denote the payoff rate at any state $s^{t+1} = (s^t, s_{t+1})$. The budget constraint is

$$g_t + R_t^b b_t = \tau_t^n w_t n_t + \tau_t^k (r_t - \delta) k_t + b_{t+1}, \tag{1.5}$$

where δ is the depreciation rate on capital.

Let $\pi_t = [\tau_t^n, \tau_t^k, R_t^b]$ denote the fiscal policy at s^t, and let $\pi = [\pi_t]$ denote the policy for all s^t. The initial stock of debt and capital, b_0 and k_0, are given.

The consumer's budget constraint is given by

$$c_t + k_{t+1} + b_{t+1} \leq (1 - \tau_t^n) w_t n_t + R_t^b b_t + R_t^k k_t, \tag{1.6}$$

where $R_t^k \triangleq 1 + (1 - \tau_t^k)(r_t - \delta)$ is the gross return on capital after taxes and depreciation.

Let $x_t = [c_t, n_t, k_{t+1}, b_{t+1}]$ denote an allocation for consumers at s^t, and let $x = [x_t]$ denote an allocation for all s^t.

Write the Lagrangian of the consumer's problem[1]:

$$\mathscr{L} \triangleq \sum_{t=0}^{\infty}\sum_{s^t}\beta^t\mu(s^t)U(c_t,n_t)+\sum_{t=0}^{\infty}\sum_{s^t}\lambda_t\big[(1-\tau_t^n)w_tn_t+R_t^bb_t+R_t^kk_t-c_t-k_{t+1}-b_{t+1}\big],$$

where $\lambda_t \equiv \lambda(s^t) \geq 0$ is the per-period multiplier. Note that $c_t \geq 0$ and $n_t \geq 0$ are implicitly required.

FOCs become

$$c_t: \quad \beta^t\mu(s^t)U_{c,t} \leq \lambda_t, \text{ w/ equality if } c_t > 0, \tag{1.7}$$

$$n_t: \quad \beta^t\mu(s^t)U_{n,t} \leq -\lambda_t(1-\tau_t^n)w_t, \text{ w/ equality if } n_t > 0, \tag{1.8}$$

$$b_{t+1}: \quad 0 = \left(\sum_{s^{t+1}}\lambda_{t+1}R_{t+1}^b - \lambda_t\right)b_{t+1}, \tag{1.9}$$

$$k_{t+1}: \quad 0 = \left(\sum_{s^{t+1}}\lambda_{t+1}R_{t+1}^k - \lambda_t\right)k_{t+1}. \tag{1.10}$$

$\forall s^\infty$, TVCs become

$$\lim \lambda_t b_{t+1} = 0, \tag{1.11}$$

$$\lim \lambda_t k_{t+1} = 0, \tag{1.12}$$

where the limits are taken over sequences of histories s^t contained in the infinite history s^∞.

Multiply (1.6) by λ_t, sum over t and s^t, rearrange to give

$$\begin{aligned}\sum_{t=0}^{\infty}\sum_{s^t}\lambda_t\big[c_t-(1-\tau_t^n)w_tn_t\big] &= \sum_{t=0}^{\infty}\sum_{s^t}\lambda_t\big(R_t^bb_t+R_t^kk_t-b_{t+1}-k_{t+1}\big)\\ &\overset{\text{(1.9) and (1.10)}}{=\!=\!=} \lambda_0(R_0^bb_0+R_0^kk_0)-\lim_{s^t\to s^\infty}\lambda_t(b_{t+1}+k_{t+1})\\ &\overset{\text{(1.11) and (1.12)}}{=\!=\!=} \lambda_0(R_0^bb_0+R_0^kk_0).\end{aligned}$$

Using (1.7) and (1.8), we can write[2]

$$\sum_{t=0}^{\infty}\sum_{s^t}\beta^t\mu(s^t)(U_{c,t}c_t+U_{n,t}n_t) = U_{c,0}(R_0^bb_0+R_0^kk_0), \tag{1.13}$$

which is the implementability constraint.

Adding (1.5) to (1.6), we obtain the resource constraint:

$$c_t+g_t+k_{t+1} \leq y_t+(1-\delta)k_t. \tag{1.14}$$

[1] By Weitzman (1973), (1.6) - (1.12) are the necessary and sufficient condition for x to solve the consumer's problem, $\forall\pi$.

[2] The interiority follows from the Inada conditions of U.

2 Planning Problem

Definition 2.1: Ramsey Equilibrium

A Ramsey equilibrium is a policy π, an allocation rule $x(\cdot)$, and price rules $w(\cdot)$ and $r(\cdot)$ s.t.

(i) π maximizes (1.4) (conditional on it) s.t. (1.6) with allocations and prices given by $x(\pi)$, $w(\pi)$, and $r(\pi)$;

(ii) $\forall \pi'$, $x(\pi')$ maximizes (1.4) s.t. (1.6) evaluated at the policy π' and the prices $w(\pi')$ and $r(\pi')$;

(iii) $\forall \pi'$, the prices satisfy (1.2) and (1.3) conditional on π'.

We adopt the convention that the initial capital tax rate τ_0^k and the initial return on debt R_0^b are fixed.

2.1 The Ramsey Allocations

It is standard to write the following proposition:

Proposition 2.2

The consumption, labor, and capital allocations $\{c_t, n_t, k_{t+1}\}$ in a Ramsey equilibrium solve the Ramsey allocation problem, that is, maximizing (1.4) s.t. implementability constraint (1.13) and resource constraint (1.14).

Using the allocations given by this proposition, we construct the bond allocation as follows.

Multiply (1.6) by λ_t, sum over all periods and states following $s^{t'}$, rearrange to give the similar result:

$$\sum_{t=t'+1}^{\infty} \sum_{s^t} \beta^{t-t'} \mu(s^t|s^{t'})(U_{c,t}c_t + U_{n,t}n_t) = U_{c,t'}(k_{t'} + b_{t'}).$$

Thus,

$$b_{t'} = \sum_{t=t'+1}^{\infty} \sum_{s^t} \frac{1}{U_{c,t'}} \beta^{t-t'} \mu(s^t|s^{t'})(U_{c,t}c_t + U_{n,t}n_t) - k_{t'}. \tag{2.1}$$

Let W incorporate the implementability constraint into the maximand, that is,

$$W(c_t, n_t, \eta) \triangleq U(c_t, n_t) + \eta(U_{c,t}c_t + U_{n,t}n_t), \tag{2.2}$$

where η is the multiplier on the implementability constraint (1.13). Thus, the Ramsey allocation problem can be represented as

$$\max \left\{ \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \mu(s^t) W(c_t, n_t, \eta) - \eta U_{c,0}(R_{k,0}k_0 + R_{b,0}b_0) \right\}$$

s.t. (1.14). Write the Lagrangian:

$$\mathscr{L} \triangleq \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \mu(s^t) W(c_t, n_t, \eta) + \sum_{t=0}^{\infty} \sum_{s^t} \theta_t [y_t + (1-\delta)k_t - c_t - g_t - k_{t+1}] - \eta U_{c,0}(R_{k,0}k_0 + R_{b,0}b_0),$$

where $\theta_t \equiv \theta(s^t) \geq 0$ is the per-period multiplier.

For $t \geq 1$, FOCs imply that

$$\begin{aligned} c_t: \quad & W_{c,t} = \theta_t, \\ n_t: \quad & W_{n,t} = -\theta_t F_{n,t}, \\ k_{t+1}: \quad & 0 = \sum_{s^{t+1}} \beta\mu(s^{t+1}|s^t)\theta_{t+1}(1-\delta+F_{k,t+1}) - \theta_t. \end{aligned}$$

That is,

$$-\frac{W_{l,t}}{W_{c,t}} = F_{n,t}, \tag{2.3}$$

$$W_{c,t} - \sum_{s^{t+1}} \beta\mu(s^{t+1}|s^t) W_{c,t+1}(1-\delta+F_{k,t+1}) = 0. \tag{2.4}$$

For $t = 0$, similarly, these conditions are

$$-\frac{W_{n,0} - \eta\left[U_{cn,0}(R_0^k k_0 + R_0^b b_0) + U_{c,0}(1-\tau_0^k)F_{kl,0}\right]}{W_{c,0} - \eta U_{cc}(R_0^k k_0 + R_0^b b_0)} = F_{n,0}, \tag{2.5}$$

$$W_{c,0} - \eta U_{cc}(R_0^k k_0 + R_0^b b_0) - \sum_{s^1} \beta\mu(s^1|s^0) W_{c,1}(1-\delta+F_{k,1}) = 0. \tag{2.6}$$

If the stochastic process on s is Markov, then (2.3) and (2.4) imply that the allocations from period 1 onward can be described by time-invariant allocation rules $c(k,s;\eta)$, $n(k,s;\eta)$, $k_+(k,s;\eta)$, and $b(k,s;\eta)$. By (2.5) and (2.6), the period 0 allocations rules are different from the stationary allocations rules that govern behavior from period 1 on.

2.2 The Ramsey Policies

In this subsection, given a Ramsey allocation (c,n,k,b) as above, we try to find the set of prices w and r, returns R^b, and tax rates τ^n and τ^k that satisfy the marginal product conditions (1.2) - (1.3), the consumer FOCs (1.7) - (1.10), and the budget constraints of the consumers and the government[3].

The prices r and w are obtained from (1.2) and (1.3). Combining them with consumer FOCs gives the optimal consumption-labor choice

$$-\frac{U_{n,t}}{U_{c,t}} = (1-\tau_t^n)F_{n,t}, \tag{2.7}$$

as well as two Euler equations[4]

$$U_{c,t} = \sum_{s^{t+1}} \beta\mu(s^{t+1}|s^t) U_{c,t+1} R_{t+1}^k, \tag{2.8}$$

$$U_{c,t} = \sum_{s^{t+1}} \beta\mu(s^{t+1}|s^t) U_{c,t+1} R_{t+1}^b. \tag{2.9}$$

The tax rate on labor τ^n is determined from (2.7), but the bond returns R^b and the capital income tax rate τ^k are indeterminate, as the following proposition shows.

[3]Since the Ramsey allocations satisfy the resource constraint, any policies and prices that satisfy the consumer budget constraint must also satisfy the government budget constraint.

[4]The capital income tax rate can be written explicitly, $\tau_t^k = 1 - \frac{R_t^k - 1}{F_{k,t} - \delta}$.

Proposition 2.3

If R^b and τ^k satisfy (1.6), (2.8), and (2.9), then so do any $\bar{R}^b$ and $\bar{\tau}^k$, where

$$\sum_{s^{t+1}} \mu(s^{t+1}|s^t) U_{c,t+1} R^b_{t+1} = \sum_{s^{t+1}} \mu(s^{t+1}|s^t) U_{c,t+1} \bar{R}^b_{t+1}, \tag{2.10}$$

$$\sum_{s^{t+1}} \mu(s^{t+1}|s^t) U_{c,t+1} \tau^k_{t+1}(F_{k,t+1} - \delta) = \sum_{s^{t+1}} \mu(s^{t+1}|s^t) U_{c,t+1} \bar{\tau}^k_{t+1}(F_{k,t+1} - \delta), \tag{2.11}$$

$$\tau^k_{t+1}(F_{k,t+1} - \delta) k_{t+1} - R^b_{t+1} b_{t+1} = \bar{\tau}^k_{t+1}(F_{k,t+1} - \delta) k_{t+1} - \bar{R}^b_{t+1} b_{t+1}. \tag{2.12}$$

Proof. See Zhu (1992). □

Remark.

Actually, there are $N-1$ degrees of indeterminacy, if in some period t, s_{t+1} can take on N values. Thus, any set of restrictions across τ^k and R^b that leads to $N-1$ restrictions at each node will lead to a unique policy. For instance, suppose that the capital tax rate cannot be contingent on the realization of the current state, i.e., for each s^t,

$$\tau^k(s^t, s_{t+1}) = \bar{\tau}^k(s^t), \ \forall s_{t+1}.$$

For another extreme, suppose that the returns on debt is not contingent on the current state, i.e., for each s^t,

$$R^b(s^t, s_{t+1}) = \bar{R}^b(s^t), \ \forall s_{t+1}.$$

While the ex post capital tax rate are not pinned down, (2.10) establishes that the ex ante tax rate is determined, which can be defined as

$$\tau^e_t \triangleq \frac{\sum_{s^{t+1}} q_{t+1} \tau^k_{t+1}(F_{k,t+1} - \delta)}{\sum_{s^{t+1}} q_t (F_{k,t+1} - \delta)}, \tag{2.13}$$

where $q_{t+1} \equiv q(s^{t+1})$ is the state price (so-called SDF) of a unit of consumption at s^{t+1} in units of consumption at s^t,

$$q_{t+1} \triangleq \beta \mu(s^{t+1}|s^t) \frac{U_{c,t+1}}{U_{c,t}}.$$

To see that clearly, we can write (2.8) as

$$1 - \sum_{s^{t+1}} q_{t+1} = \sum_{s^{t+1}} q_{t+1}(F_{k,t+1} - \delta) - \sum_{s^{t+1}} q_{t+1} \tau^k_{t+1}(F_{k,t+1} - \delta)$$

Thus,

$$\tau^e_t = 1 - \frac{1 - \sum_{s^{t+1}} q_{t+1}}{\sum_{s^{t+1}} q_{t+1}(F_{k,t+1} - \delta)} = \frac{1 + \sum_{s^{t+1}} q_{t+1}(F_{k,t+1} - \delta - 1)}{\sum_{s^{t+1}} q_{t+1}(F_{k,t+1} - \delta)},$$

which does not depend on the ex post tax rate τ^k_t.

Similarly, given (2.12), the revenues from capital taxation minus the value of outstanding debt is determined. Imagine that the government achieves the desired state contingency in debt returns by promising a state-non-contingent rate of return on government debt $\bar{r}_t$ that satisfies

$$\sum_{s^{t+1}} q_{t+1} R^b_{t+1} = \sum_{s^{t+1}} q_{t+1}(1 + \bar{r}_t), \tag{2.14}$$

and by levying a state-contingent tax τ^b_{t+1} on interest payments from government debt that satisfies

$$R^b_{t+1} = 1 + \bar{r}_t(1 - \tau^b_{t+1}). \tag{2.15}$$

From (2.9), we can write the state-non-contingent rate as

$$\bar{r}_t = \frac{1 - \sum_{s^{t+1}} q_{t+1}}{\sum_{s^{t+1}} q_{t+1}}.$$

From (2.1), b_t is pinned down. Thus, (2.12) can be thought of as pinning down the sum of the tax revenues from the capital income tax and the debt income tax given by

$$\tau^k_{t+1}(F_{k,t+1} - \delta)k_{t+1} + \tau^b_{t+1}\bar{r}_t b_{t+1} = (1 + \bar{r}_t)b_{t+1} + \text{constant}.$$

We transform these revenues into a rate by dividing them by the income from capital and debt to obtain

$$\tau^a_{t+1} \triangleq \frac{\tau^k_{t+1}(F_{k,t+1} - \delta)k_{t+1} + \tau^b_{t+1}\bar{r}_t b_{t+1}}{(F_{k,t+1} - \delta)k_{t+1} + \bar{r}_t b_{t+1}}. \tag{2.16}$$

All (certain) tax rates can be described by time-invariant policy rules of the form $\tau^n(k, s; \eta)$, $\tau^e(k, s; \eta)$, and $\tau^a(k, s; \eta)$ from period 1 on. In contract, the policy rules for period 0 are different from these time-invariant rules.

3 Computation Strategy

3.1 Calibration

The preferences is chosen as King, Plosser, and Rebelo (1988):

$$U(c, n) = \frac{\left(c^\gamma (1 - n)^{1-\gamma}\right)^{1-\sigma}}{1 - \sigma}.$$

The technology takes the form as

$$y_t = k_t^\alpha [n \cdot \exp(\rho t + z_{t+1})]^{1-\alpha},$$

where z_t follows an AR(1) process,

$$z_{t+1} = \rho_z z_t + \varepsilon^z_t, \quad \varepsilon^z_t \sim \mathcal{N}(0, \sigma^2_z).$$

Let government spending be given by

$$g_t = G \cdot \exp(\rho t + \tilde{g}_t),$$

where $\tilde{g}_t$ follows an AR(1) process,

$$\tilde{g}_{t+1} = \rho_g \tilde{g}_t + \varepsilon^z_t, \quad \varepsilon^g_t \sim \mathcal{N}(0, \sigma^2_g).$$

The baseline model sets $\sigma = 1$, that is, $U = \gamma \ln c + (1 - \gamma)\ln(1 - n)$. We also consider a high risk aversion model for $\sigma = 9$. Using the same procedure as in Christiano and Eichenbaum (1992), we firstly consider a non-stochastic stationary economy to take parameterization.

We choose the capital-output ratio, the fraction of available time worked, the ratio of government spending to output, and the debt-output ratio as 2.71, 0.23, 0.18, and 0.51, respectively, to match the empirical estimation in the U.S.

The values of the capital and labor tax rates are 27.1 and 23.7 percent, respectively, which is referred as the non-stochastic benchmark policies. The initial conditions for our experiments are given by the BGP of a deterministic economy with these policies, $k_0 = 1.05$ and $R^b_0 b_0 = 0.20$.

Other parameters are chosen as follows:

Table 1: Baseline Model Parameter Values

Preferences	$\gamma = .25$	$\sigma = 1$	$\beta = .98$
Technology	$\alpha = .34$	$\delta = .08$	$\rho = .016$
Stochastic process for government consumption	$G = .07$	$\rho_g = .89$	$\sigma_g = .07$
Stochastic process for technology shock	$\rho_z = .81$	$\sigma_z = .04$	

3.2 Numerical Procedure

We briefly describe our numerical procedure as follows.

3.2.1 Step 1. De-trend

We use the standard procedure of transforming our economy with growth into one without growth. This transformation affects only the discount factor and the depreciation rate.

3.2.2 Step 2. Compute the Ramsey allocations with fixed multiplier on IC

Guess a initial value of η, the multiplier on the implementability constraint. Given this value of η, the solutions to the Ramsey allocation problem for $t \geq 1$ are stationary functions of (k, s), where $s \triangleq (z, \tilde{g})$.

For $t \geq 1$, we use the resource constraint (1.14) and the intratemporal condition (2.3) to express $c(k, s)$ and $n(k, s)$ in terms of k, s, and $k'(k, s)$. We use the Euler equation (2.4) to determine $k'(k, s)$[5].

For $t = 0$, we use the resource constraint (1.14) and the intratemporal condition (2.5) to express c_0 and n_0 in terms of k_1, and the Euler equation (2.6) to determine k_1.

For $t \geq 1$, (2.1) establishes that the bond allocation rule is also a stationary function of (k, s), given η. The bond allocation can be recursively written as

$$U_c(\bar{s})b(\bar{s}) = \sum_{s'} \beta\mu(s'|s)\left[U_c(\bar{s}')c(\bar{s}') + U_n(\bar{s}')n(\bar{s}') + U_c(\bar{s}')b(\bar{s}') + U_c(\bar{s}')k'(\bar{s}')\right] - U_c(\bar{s})k'(\bar{s}), \tag{3.1}$$

where $\bar{s} \equiv (k, s)$. The stationary bond allocation rule is the fixed point of the mapping $b \to b'$ implied by this equation[6].

For $t = 0$, we substitute the period 0 consumption and employment rules into the marginal utility on the LHS of (3.1) and the stationary rules on the RHS to derive $b_1(\cdot)$.

3.2.3 Step 3. Solve for the multiplier on IC

We substitute $b_1(\cdot)$ with the other period 0 allocation rules into the consumer budget constraint (1.6) evaluated in period 0.

Combining (1.7) with (1.8) gives that $U_{c,t}/U_{n,t} = -(1 - \tau_t^n)w_t$. Using this equation and setting $\tau_0^k = 27.1\%$ (s_0 is pre-determined), we calculate a value for $R_0^b b_0$.

Finally, we iterate on η until $R_0^b b_0 = 0.20$ is satisfied.

3.2.4 Step 4. Compute the Ramsey policies

We compute the tax rate on labor income, the ex ante tax on capital income, and the tax on private assets by substituting the allocation rule into (2.7), (2.13), and (2.16), respectively.

[5] Finding a function $k'(k, s)$ that satisfies (2.4), $\forall k, s$, is computationally infeasible. Thus we limit ourselves to a finite-parameter class of decision rules, see the technical appendix of this paper.

[6] Finding a function $b(k, s)$ that satisfies (3.1), $\forall k, s$, is computationally infeasible. Instead, we restrict the bond rule to be continuous and piecewise linear in k for each fixed s and require that (3.1) be satisfied at a finite set of points, see the technical appendix of this paper.